Hari Patel

267-905-5972 | Philadelphia, PA | haririteshpatel@gmail.com | linkedin.com/in/haripatel788 | haripatel788.github.io

Education

Drexel University | Pennoni Honors College — Philadelphia, PA

B.S. Computer Science; Minor in Artificial Intelligence & Machine Learning — *Expected Jun. 2030*

- **GPA:** 3.88/4.00 **Honors:** Presidential Scholarship; Dean's List (Fall, Winter, Spring 2025–26)
- **Coursework:** Computer Programming I & II, Software Engineering, Multivariate Calculus, Linear Algebra

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, FastAPI, Express.js, Tailwind CSS, Scikit-Learn, Pandas, NumPy
Tools & Databases: Git, GitHub, PostgreSQL, Supabase, Firebase, Vercel, AWS (EC2, RDS), pytest
Concepts: Data Structures & Algorithms, OOP, REST API Design, Database Design, Authentication & Authorization, Unit Testing, Machine Learning, LLM Integration
Practices: Full-Stack Development, Agile, Code Review

Experience

Undergraduate Researcher | *Drexel University (Advisor: Dr. Arvin Ebrahimkhanlou)* — Oct. 2026 – Present

- Reviewing CVPR literature on generative AI for causal reasoning in structural condition diagnosis; surveying candidate architectures for a multimodal pipeline.

Co-Founder, Novus | *React, TypeScript, Supabase, PostgreSQL, Llama 3.3 70B* — Apr. 2026 – Present

- Built a multi-tenant backend on Supabase with a 40-table PostgreSQL schema, using Row-Level Security (RLS) to isolate each institution's data and enforce role-based access control.
- Built an LLM grading pipeline (Llama 3.3 70B via Supabase Edge Functions) that scores answers on technical accuracy, risk awareness, and communication; its scores matched my manual grades within 2 points on 37 of 40 test answers.
- Wrote 80+ technical interview questions and reviewed them with 8+ finance industry advisors; built real-time scoring endpoints for the beta release.

Software Engineering Mentee, CodeHS — Jun. 2025 – Sep. 2025

- Sat in on sprint ceremonies and reviewed pull requests and tickets while mentored by an 8-person production engineering team.

Projects

Market Pulse | *Python, FastAPI, Pandas, Supabase* — Feb. 2026 – Present

- Built a REST API that forecasts stock volatility with per-stock Heterogeneous Autoregressive (HAR) models trained on 10 years of price history, returning expected daily moves and 80%/95% prediction intervals.
- Added Supabase JWT authentication, tiered RBAC, sliding-window rate limiting, and 27 pytest integration tests.
- Backtested 112,700 predictions across 490 S&P 500 equities; beat an EWMA baseline on 86% of 474 held-out stocks at 1-, 5-, and 20-day horizons, with 80% prediction intervals covering 81% of outcomes.

Sixth & Market | *React, TypeScript, Express.js, Firebase, Groq API, Leaflet* — Apr. 2026

- Placed 9th of 52 teams at Philly Codefest 2026 with a civic analytics app built and deployed in an 11-hour hackathon.
- Used Groq LLM endpoints behind an Express.js reverse proxy to classify Philadelphia 311 service tickets, keeping API keys server-side.
- Built an interactive Leaflet map combining FCC broadband coverage, U.S. Census tracts, and municipal complaint data to show connectivity dead zones.

PrepGo | *Next.js, TypeScript, Tailwind CSS, PostgreSQL (Neon), Clerk, Vercel* — Jan. 2026 – Mar. 2026

- Deployed a full-stack SAT prep platform with Clerk authentication; 30 active users completed 300+ practice questions.
- Added a per-session rate limiter that fixed server crashes caused by database connection pool exhaustion.
- Built the Next.js/Tailwind CSS frontend and the backend queries against a serverless Neon PostgreSQL database.

Leadership & Volunteering

Youth Coordinator — Sep. 2025 – Present

Bochasanwasi Shri Akshar Purushottam Sanstha — *Philadelphia, PA*

- Ran weekly programs for high-school and college youth; average attendance grew from 14 to 19 over 52 weeks.
- Organized 7 community events and did personal outreach to bring newcomers into the group.